Genes and Memes

John Maynard Smith in The London Review of Books, *4–18 February 1982. (Extract from review of* The Extended Phenotype.)

The Selfish Gene was unusual in that, although written as a popular account, it made an original contribution to biology. Further, the contribution itself was of an unusual kind. Unlike David Lack's classic *The Life of the Robin*—also an original contribution in popular form—*The Selfish Gene* reports no new facts. Nor does it contain any new mathematical models—indeed it contains no mathematics at all. What it does offer is a new world view.

Although the book has been widely read and enjoyed, it has also aroused strong hostility. Much of this hostility arises, I believe, from misunderstanding, or rather, from several misunderstandings. Of these, the most fundamental is a failure to understand what the book is about. It is a book about the evolutionary process—it is not about morals, or about politics, or about the human sciences. If you are not interested in how evolution came about, and cannot conceive how anyone could be seriously concerned about anything other than human affairs, then do not read it: it will only make you needlessly angry.

Assuming, however, that you are interested in evolution, a good way to understand what Dawkins is up to is to grasp the nature of the debates which were going on between evolutionary biologists during the 1960s and 1970s. These concerned two related topics, 'group selection' and 'kin selection'. The 'group selection' debate was sparked off by Wynne-Edwards [who suggested that behavioural adaptations] had evolved by 'group selection'—i.e. through the survival of some groups and the extinction of others …

At almost the same time, W. D. Hamilton raised another question about how natural selection acts. He pointed out that if a gene were to cause its possessor to sacrifice its life in order to save the lives of several relatives, there might be more copies of the gene present afterwards than if the sacrifice had not been made … To model the process quantitatively, Hamilton introduced the concept of 'inclusive fitness' … which includes, not only an individual's own offspring, but any additional offspring raised by relatives with the help of that individual, appropriately scaled by the degree of relationship… .

Dawkins, while acknowledging the debt we owe to Hamilton, suggests that he erred in making a last-ditch attempt to retain the concept of fitness, and that he would have been wiser to adopt a full-blooded 'gene's eye' view of evolution. He urges us to recognise the fundamental distinction between 'replicators'—entities whose precise structure is replicated in the process of reproduction—and 'vehicles': entities which are mortal and which are not replicated, but whose properties are influenced by replicators. The main replicators with which we are familiar are nucleic acid molecules—typically DNA molecules—of which genes and chromosomes are composed. Typical vehicles are the bodies of dogs, fruitflies and people. Suppose, then, that we observe a structure such as the eye, which is manifestly adapted for seeing. We might reasonably ask for whose benefit the eye has evolved. The only reasonable answer, Dawkins suggests, is that it has evolved for the benefit of the replicators responsible for its development. Although like me, he greatly prefers individual to group advantage as an explanation, he would prefer to think only of replicator advantage.

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